



WIESŁAW PIETRUSZKIEWICZ

USER EXPERIENCE

UX – Which one do you prefer?



UX Examples #1



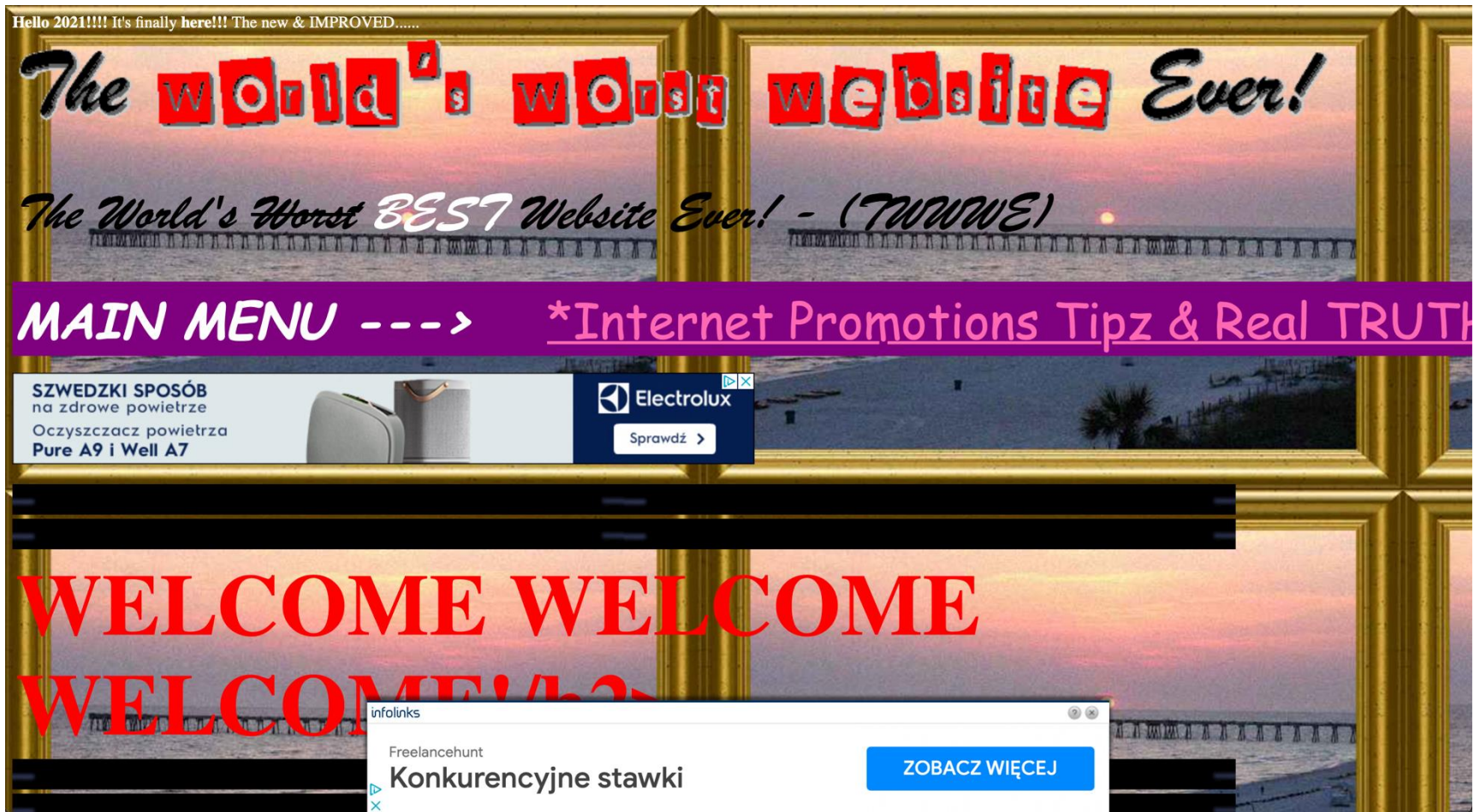
UX Examples #2



UX Examples #3



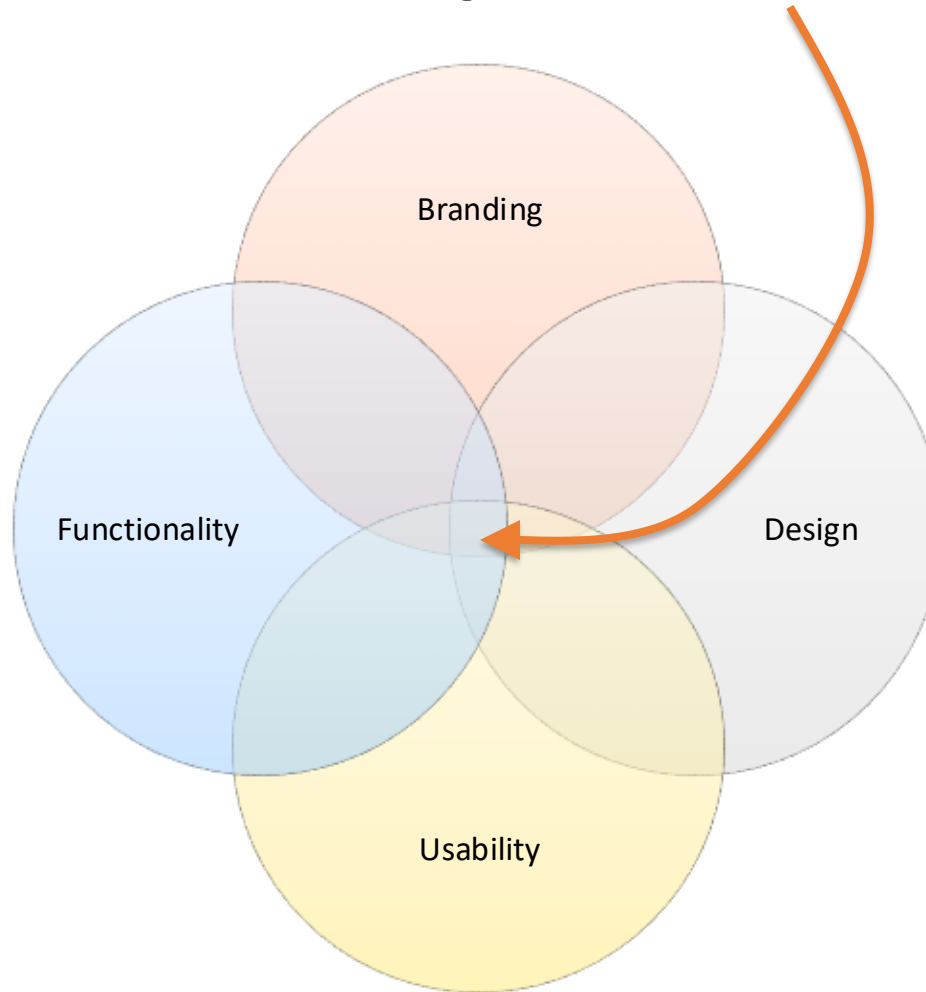
The world's worst website ever ...



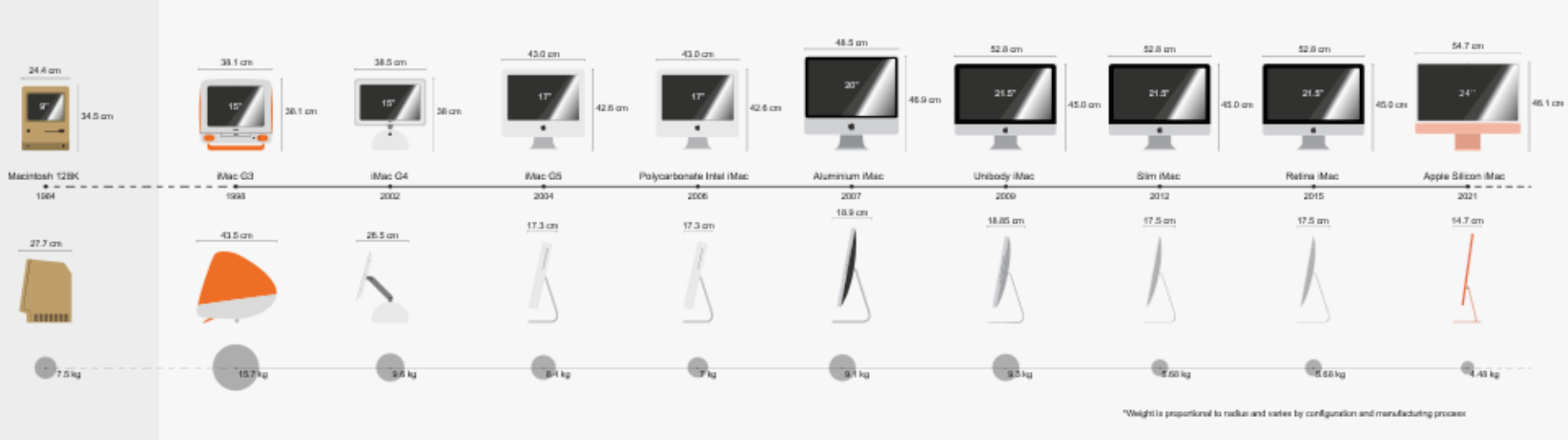
Meaning of UX in Web Design

- User Experience – any interaction a user makes with service or product
- A transition of focus - from what can be done using designed system to how it will be done
- There are two approaches to this subject:
 - General principles inducted from multiple cases
 - Tested improvements (A/B testing e.g. see <https://netflixtechblog.com>)

User Experience



An example of UX focused design



Timeline of the product Apple iMac, Giulia Piccoli Trapletti, DensityDesign Research Lab

Hick's Law



15 Rules of UX by Adobe Inc #1

UX is not (only) UI

- User interface is only a part of UI; UX is more a general emotional outcome;

Know your audience

- Perform user research and provide benefits rather than features;

You are not the user

- Don't assume the other share your beliefs and behaviour; Test the real users;

Adapt design for short attention spans

- The average attention span is limited (8 seconds - as researched by Microsoft);

The UX process isn't set in stone

- Adapt process for the product;

15 Rules of UX by Adobe Inc #2

Prototype before you build a real product

- Prototyping helps deliver proper design;

Use real content when designing

- Avoid fake test and images;

Keep things simple and consistent

- The intuition is a first source of actions, not the manuals;

Recognition over recall

- Self explanatory systems does not require to recall essential information;

Make design usable and accessible

- Keep in mind a diverse range of users;

15 Rules of UX by Adobe Inc #3

Don't try to solve a problem yourself

- Cooperate with other team members and let them evaluate your work;

Don't try to solve everything at once

- Use iterative process of improvements;

Preventing errors is better than fixing them

- The cost of prevention is significantly lower than cost of fixation;

Offer informative feedback

- Interact with user and provide feedback;

Avoid dramatic redesigns

- Introduce small changes more frequently than large ones rarely;

Progressive disclosure

- A design pattern in which information displayed in a minimal range, necessary at a given moment
- Typical UX application is called Responsive (Staged) Disclosure – forms are constructed on a basis of previous selections

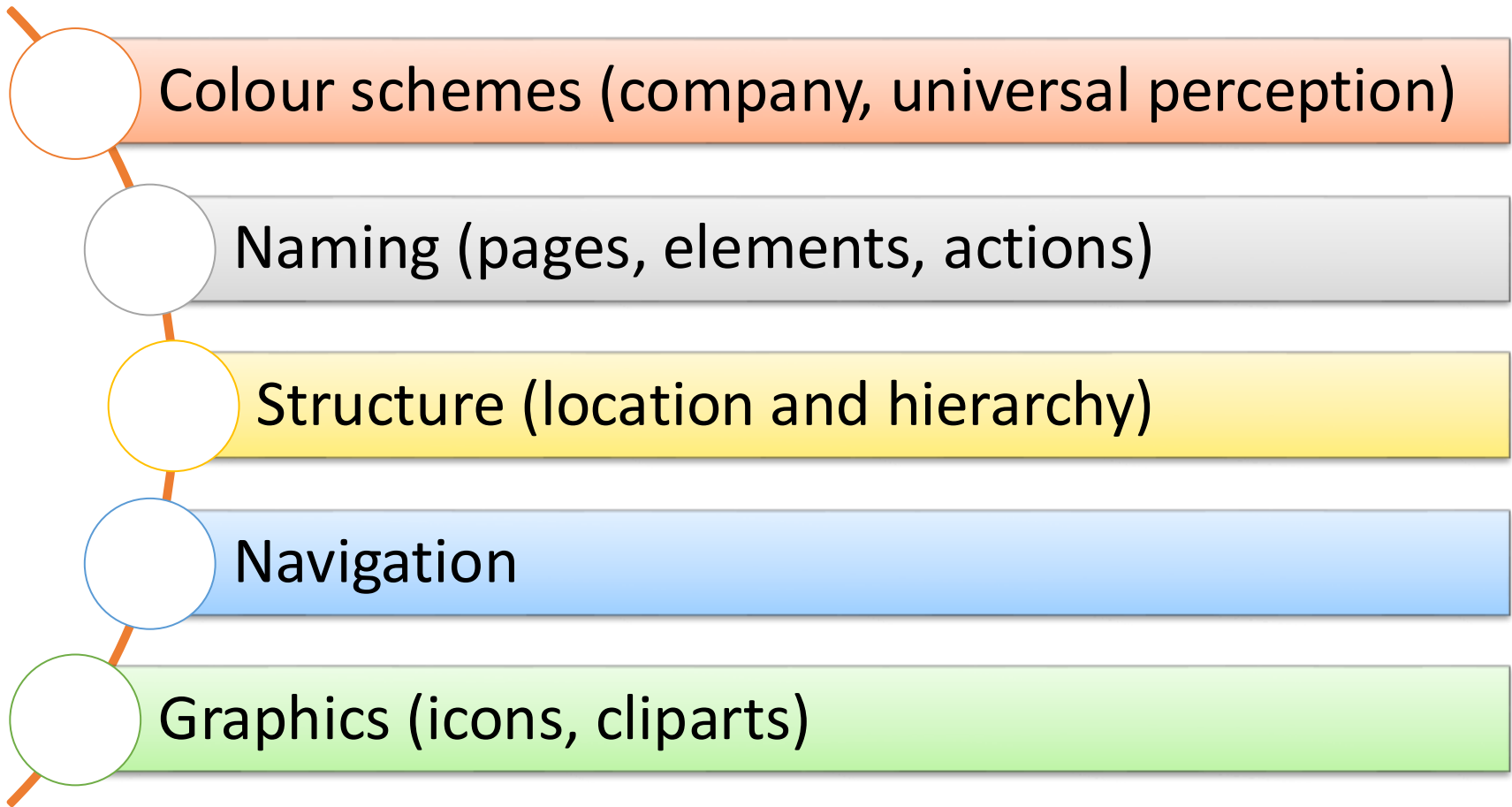


Empty state / blank slate

- No data to display may confuse the users
- Adding a content may cause UX to improve
- Possible information that could be added:
 - Tutorials
 - Screenshots / demos
 - Checklists encouraging the users
 - Welcome messages, with explanation of values created by your services



Conventions



Affordances

Meaning of affordance

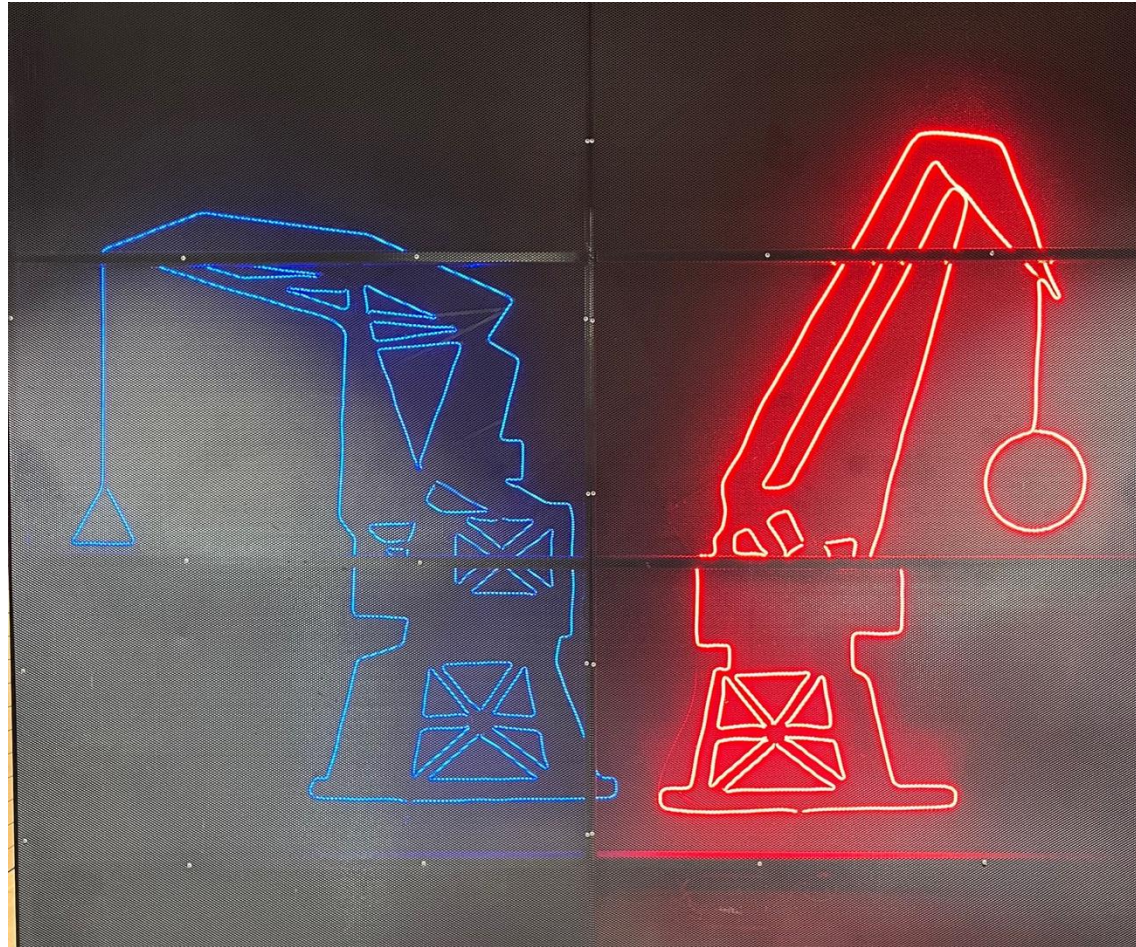
- Affordance is an attribute of an object presenting a prompt on what can be done with it
- For web it can be in form of:
 - Visuals (photos, signs, icons, UI fields)
 - Patterns (usage habits)
 - Copies (textual slogans)



Symbolics

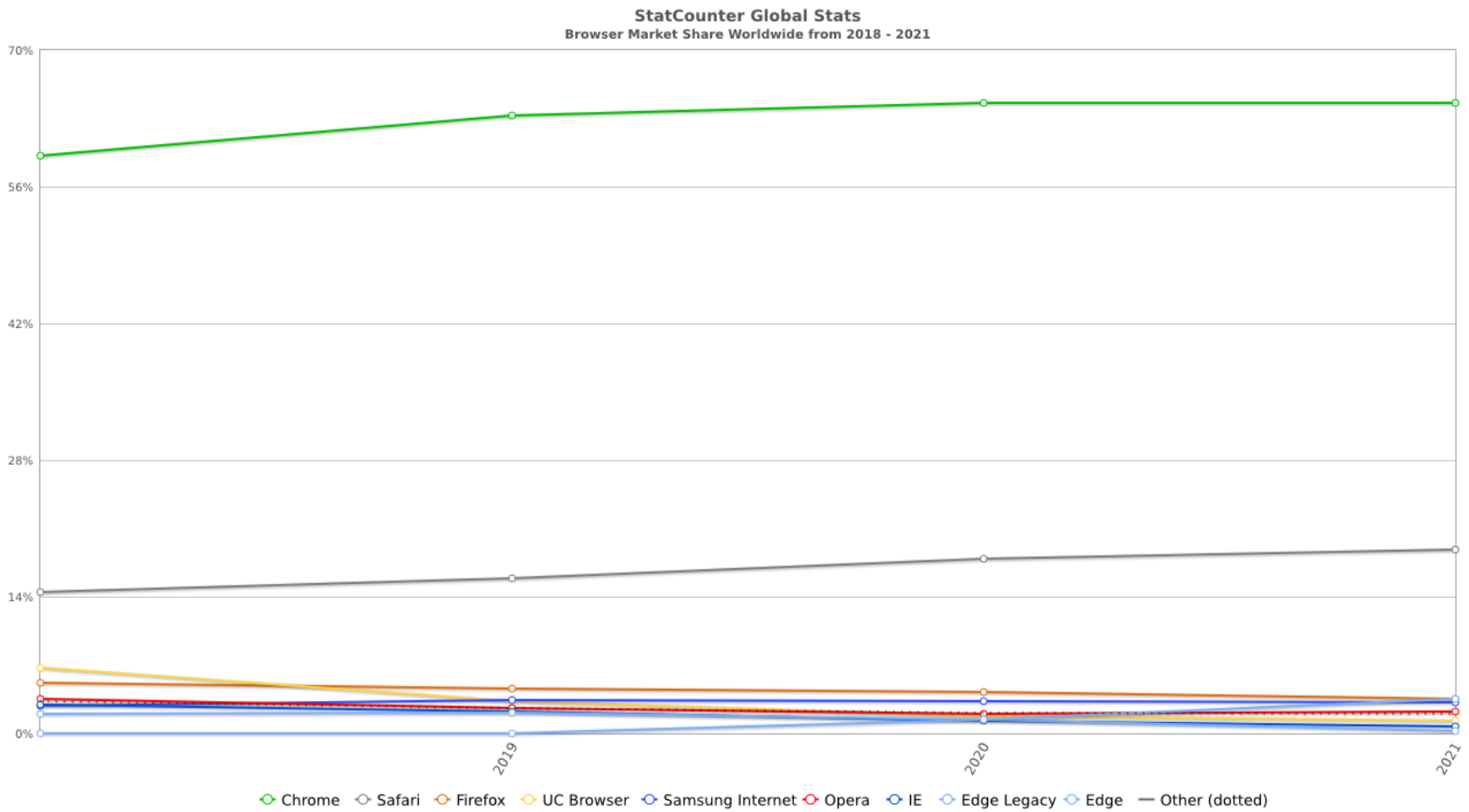


A local example

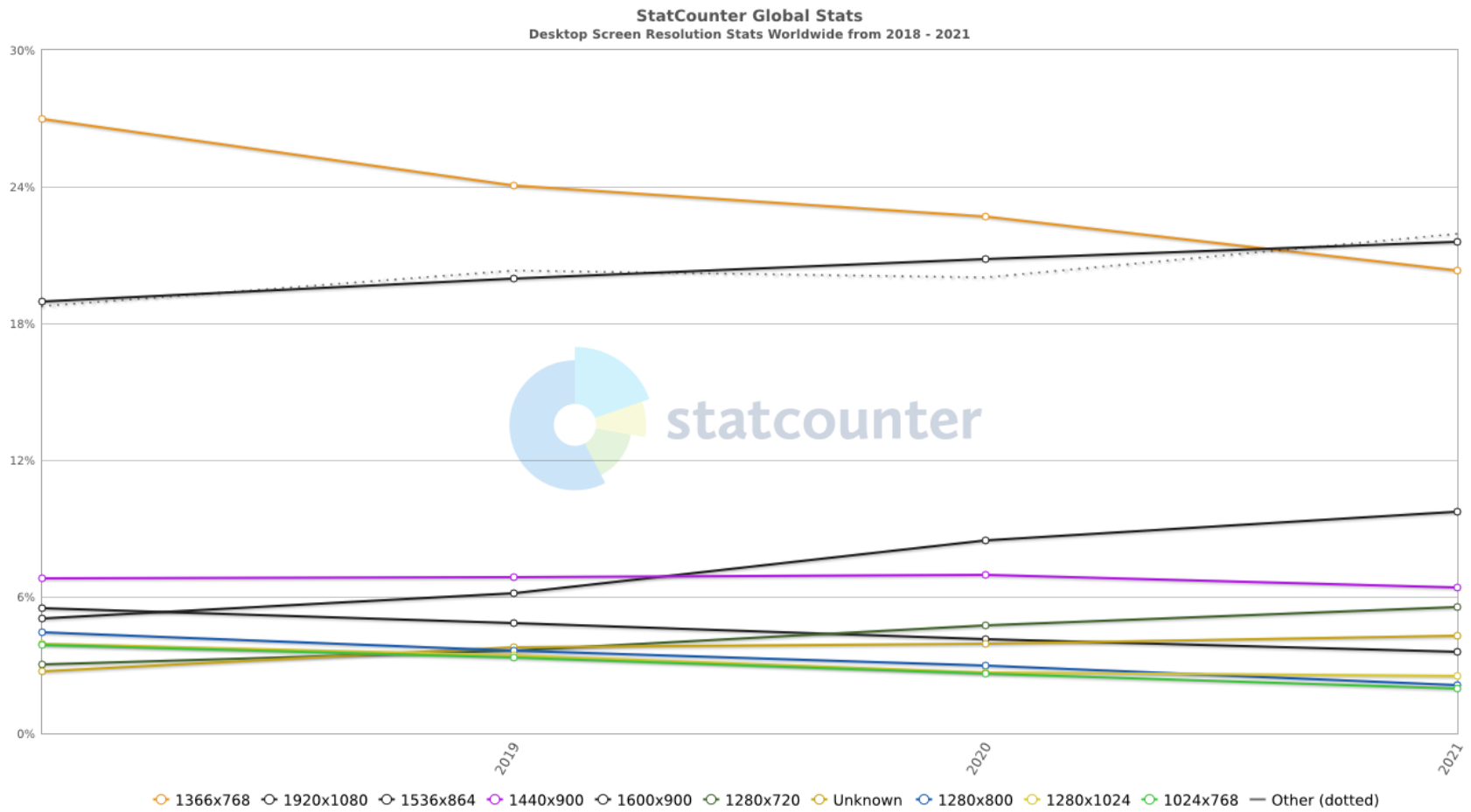


WEB EVALUATION – USER'S PERSPECTIVE

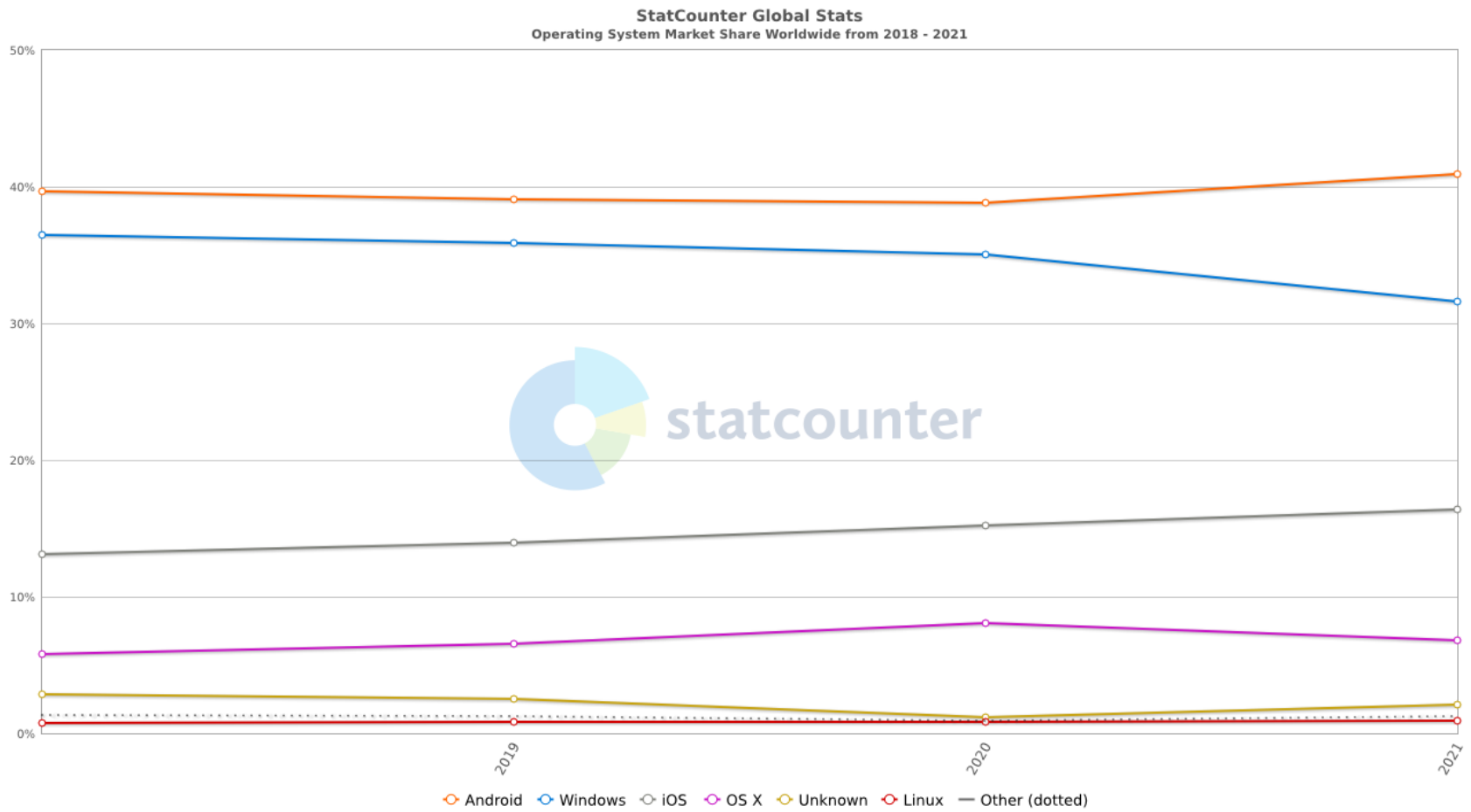
Web Design Key Factors – Browser



Web Design Key Factors – Resolution



Web Design Key Factors – OS



Web Design Key Factors – Extensions

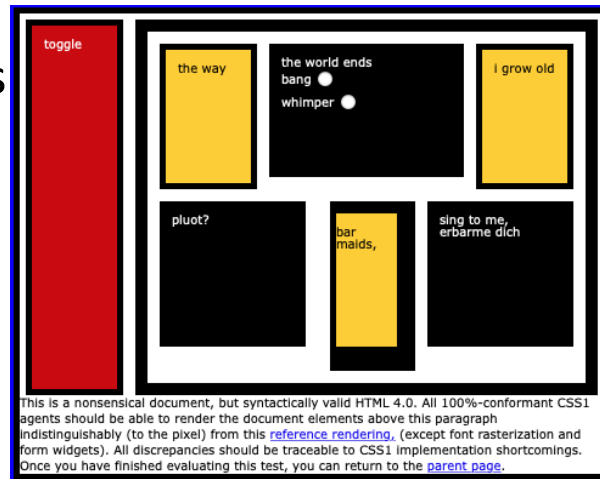
- Remember about browser extensions as they may influence how site will be displayed:
 - Ad blockers (e.g. no ads – no revenue)
 - Tracker blocking (e.g. no session or analytics)
 - Extra visual addons (e.g. broken structure)
- Test web site on a *vanilla* browser
- If necessary use extension detection and blocking (e.g. counter ad-blocking)

AcidTests

Acid Tests

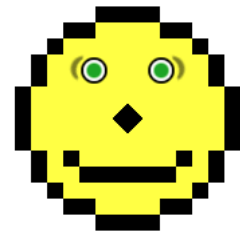
- Verify if a test page rendered by browser matches the reference
- Standardisation increases interoperability of web-based solutions
- Avoid non-standard solutions (technologies, browsers)
- A non-standard proposal may become a standard in the future

AcidTest1



AcidTest2

Hello World!

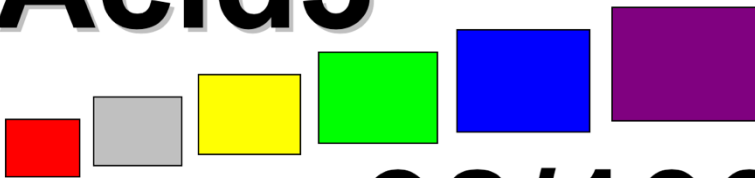


Acid3

Safari 15.5 on MacOS

Reference image

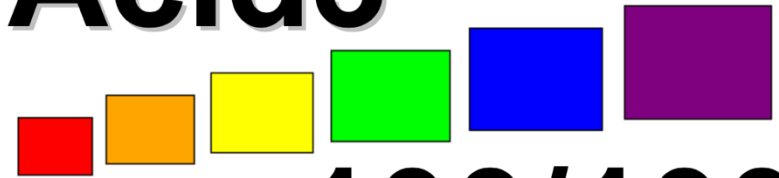
Acid3



98/100

To pass the test, a browser must use its default settings, the animation has to be smooth, the score has to end on 100/100, and the final page has to look exactly, pixel for pixel, like [this reference rendering](#).

Acid3



100/100

To pass the test, a browser must use its default settings, the animation has to be smooth, the score has to end on 100/100, and the final page has to look exactly, pixel for pixel, like [this reference rendering](#).

HTML5Test

The image displays two screenshots of the HTML5Test website, which evaluates browser support for HTML5 features. Both screenshots show a score out of 555 points.

Top Screenshot (Safari 15.1):

- Header: HTML5TEST how well does your browser support html5?
- Navigation: your browser (selected), other browsers, compare, news, device lab, about the test
- Score: YOUR BROWSER SCORES **515** OUT OF 555 POINTS
- Browser Info: You are using Safari 15.1 on macOS Catalina 10.15
- Buttons: Save results, Compare to..., Share, Donate
- Footer: SERIOUS about website SCREENSHOTS? Try URLBOX

Bottom Screenshot (Chrome Dev 95.0.4638.69):

- Header: HTML5TEST how well does your browser support html5?
- Navigation: your browser (selected), other browsers, compare, news, device lab, about the test
- Score: YOUR BROWSER SCORES **528** OUT OF 555 POINTS
- Browser Info: You are using Chrome Dev 95.0.4638.69 on macOS Catalina 10.15
- Buttons: Save results, Compare to..., Share, Donate
- Footer: SERIOUS about website SCREENSHOTS? Try URLBOX

Responsiveness #1

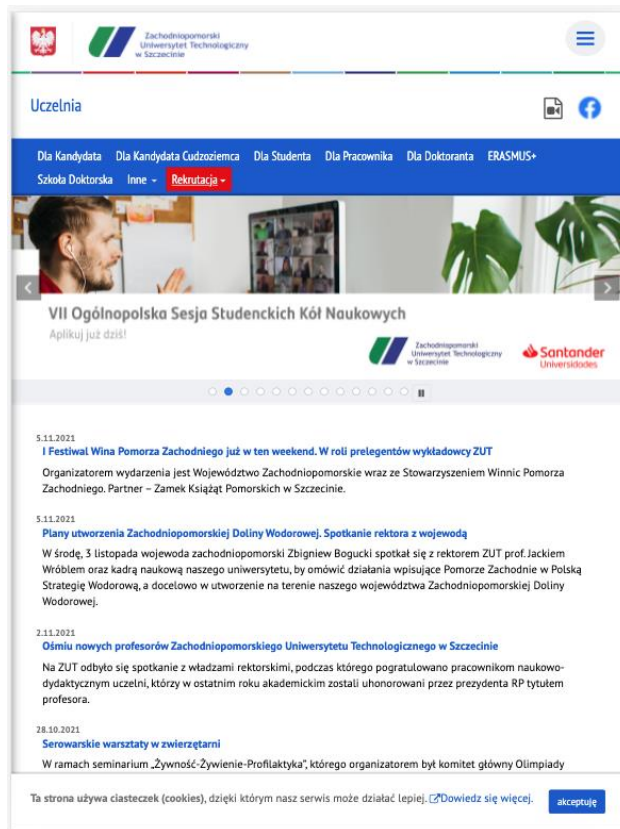
- Responsive web design – an approach to web design to make them render on various devices efficiently using displays, providing high usability and satisfaction
- Common approach - Fluid grids, flexible images & CSS media queries
- A single page with responsive layout vs multiple (typically – desktop&mobile) layouts

Responsiveness #2

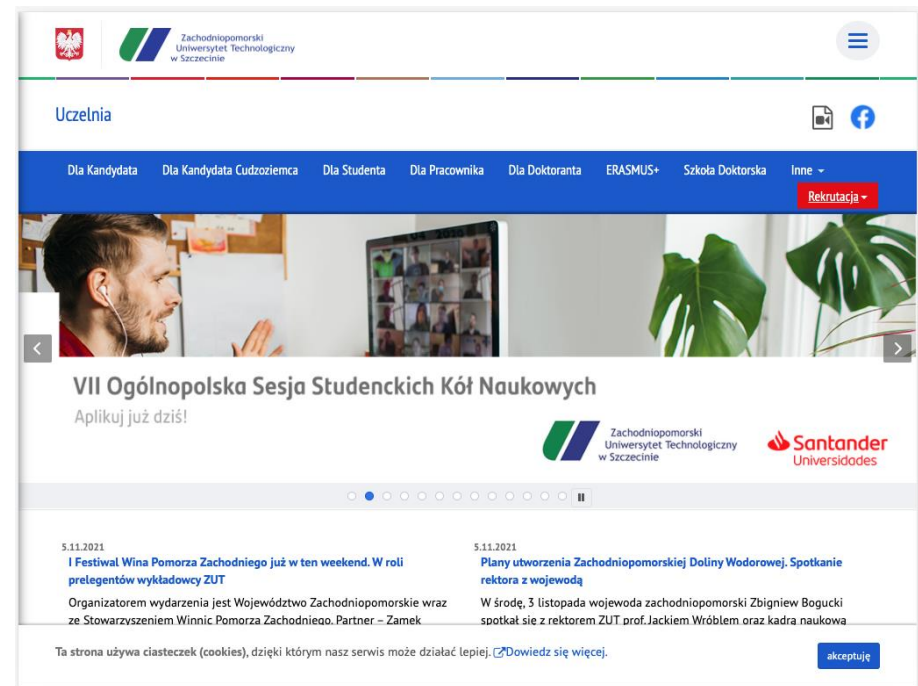
- Responsiveness – a value provided by popular front-end templates
- Simplifies the development and testing of website (assuming focus on multiple devices)
- Requires the website design to include planning how the content will be presented
- Main benefit – mobile-friendly web pages
- + Search engine boosting for the above

Responsiveness – An example #1

Simulated iPad

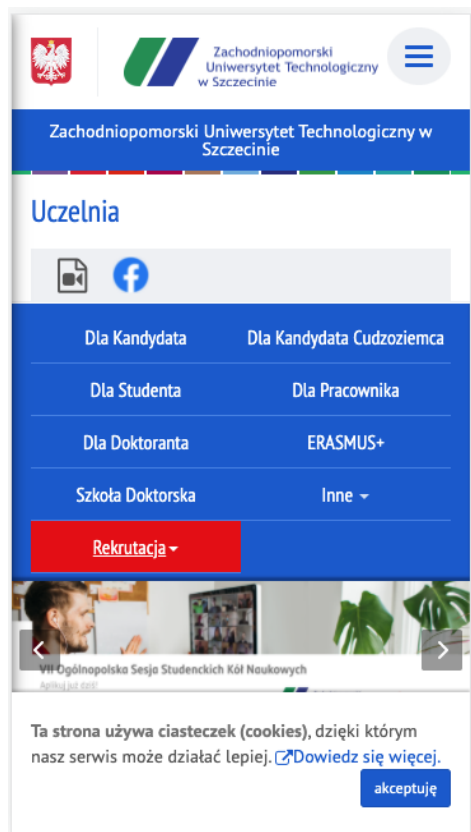


Simulated rotated iPad

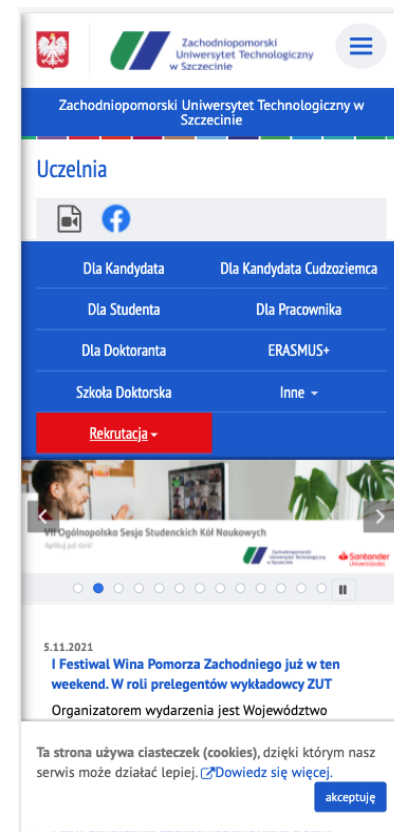


Responsiveness – An example #2

Simulated Galaxy S5



Simulated iPhone X

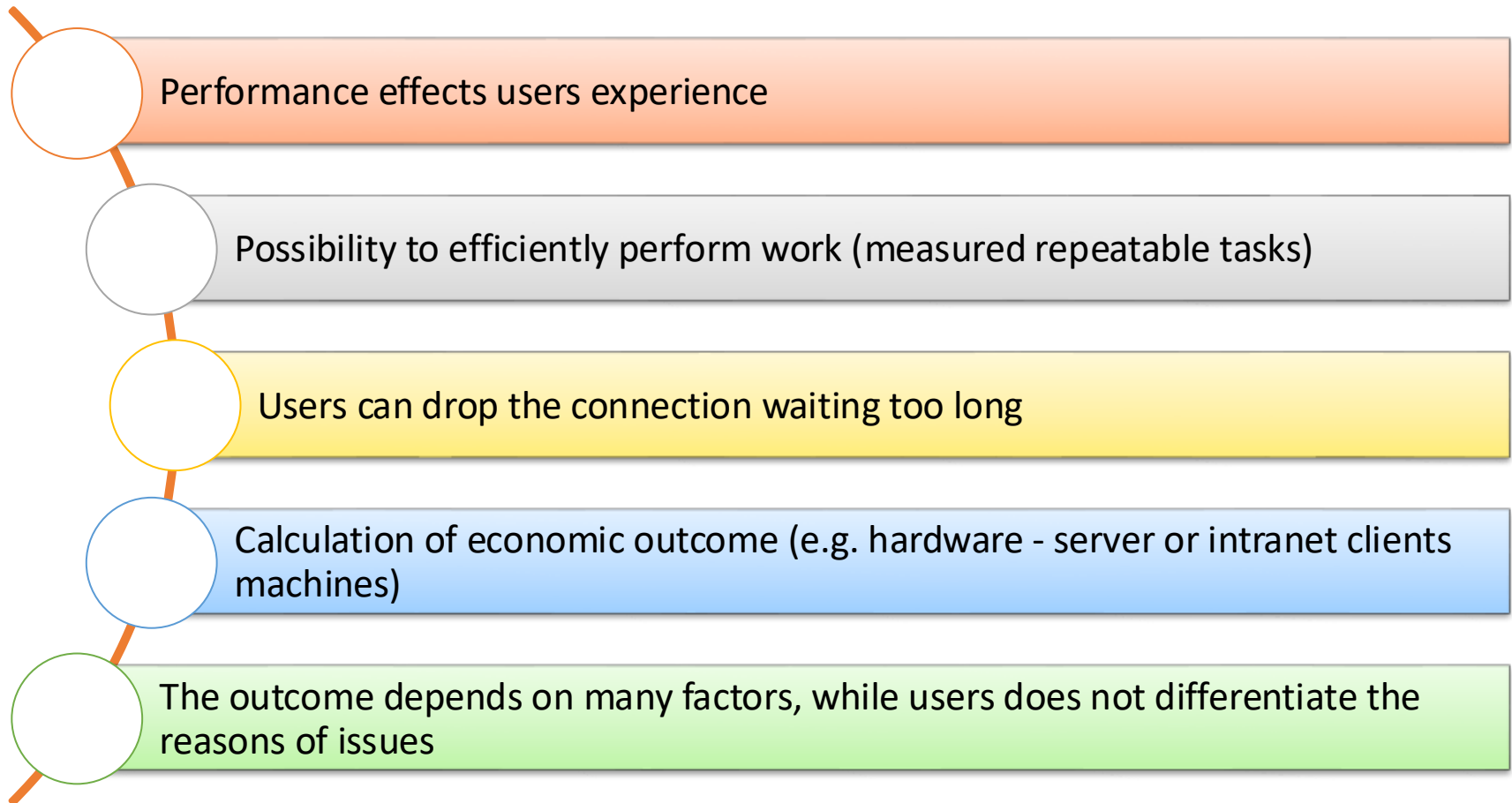


Special CSSes

- For public organisations it became a common practice to provide special CSSes (high contrast, safe colours, increased font size)
- For business entities provision of special CSSes increases the audience
- Many organisations use a single version of page constructed with vision issues in mind

WEB EVALUATION – TECHNOLOGICAL PERSPECTIVE

Performance – General



The performance analysis – Dependencies

Speed of loading

- Size of data to download
- Number of items
- Speed of connection
- Server's performance
- Server's load
- Download contains (enforced sequential order)
- Presence of cache

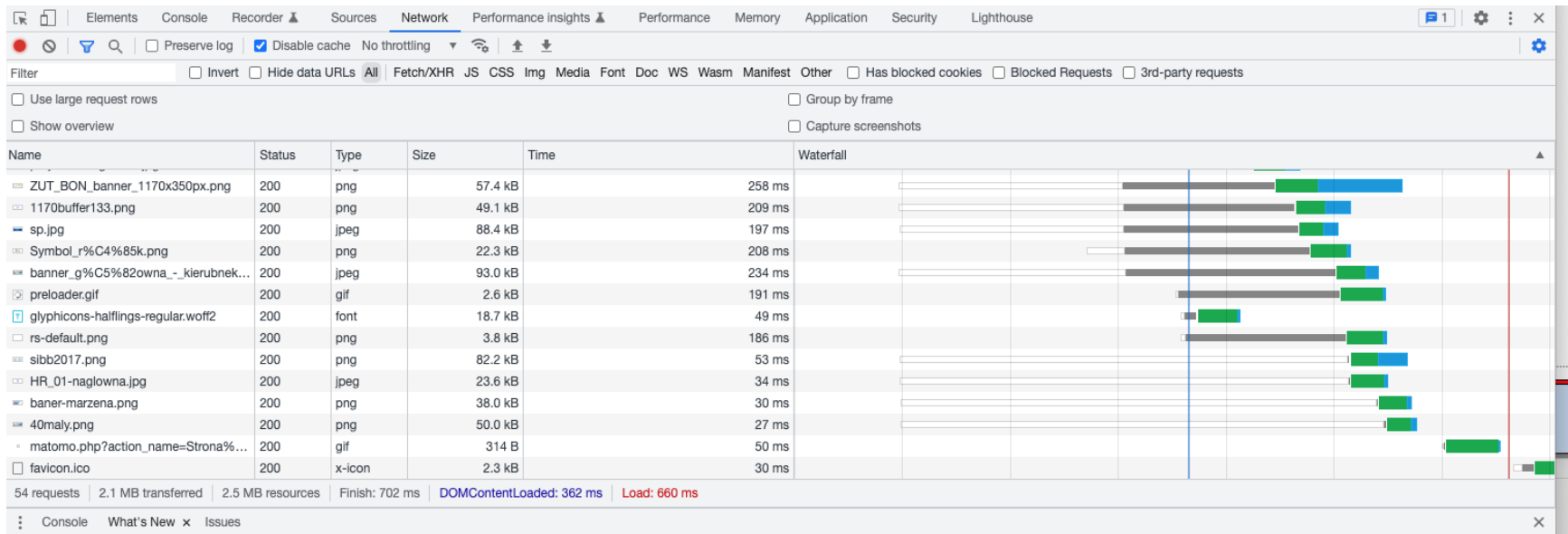
Speed of rendering

- Type of elements
- Number of elements
- Availability at a given time
- Interdependencies of elements (order of rendering)
- Necessity to calculate dimensions
- Partial rendering vs full load rendering

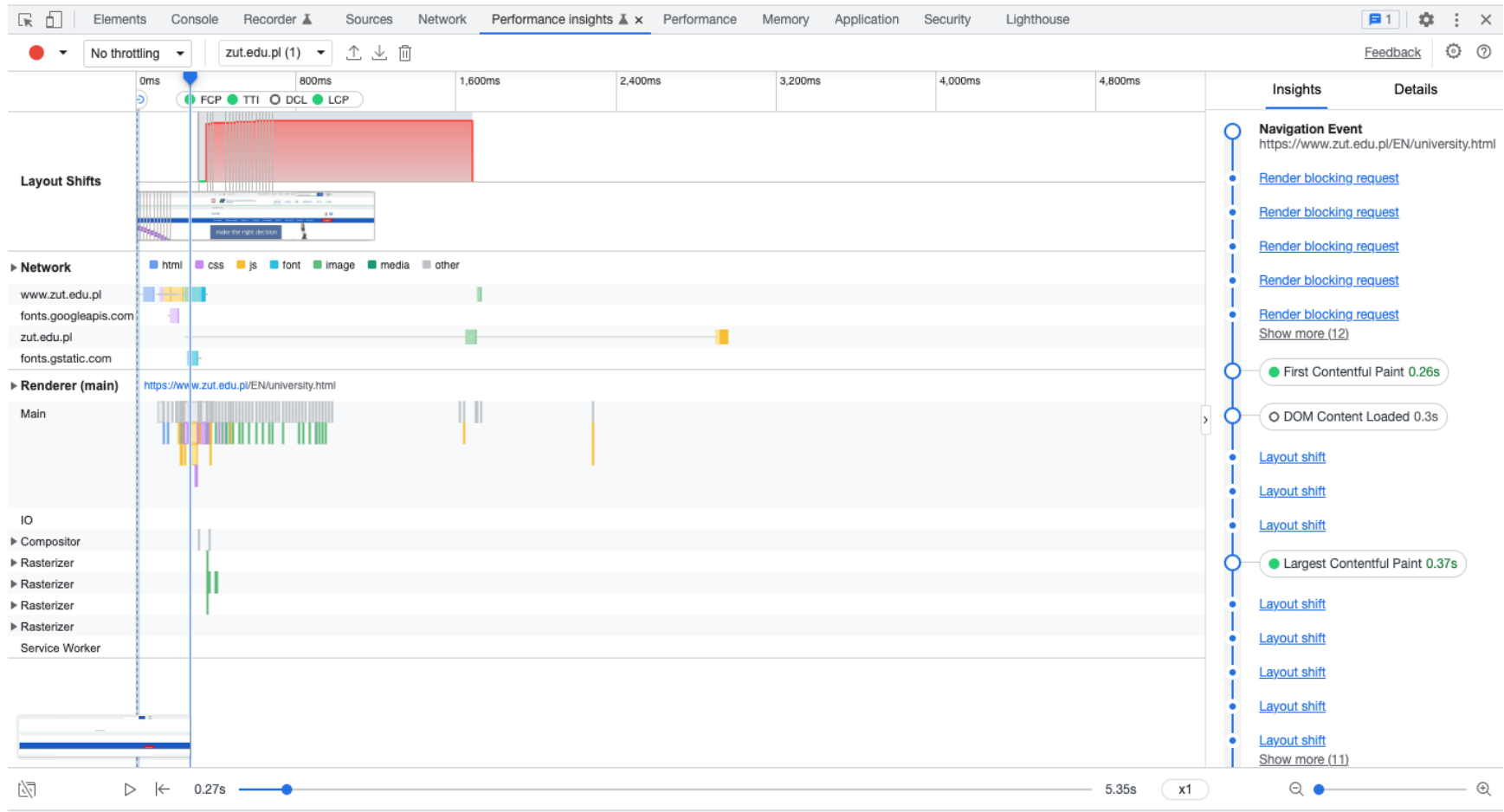
Memory usage

- Type of elements
- Size of elements
- Memory garbage collection/cleansing
- Code optimisation
- External dependencies (JS libraries)

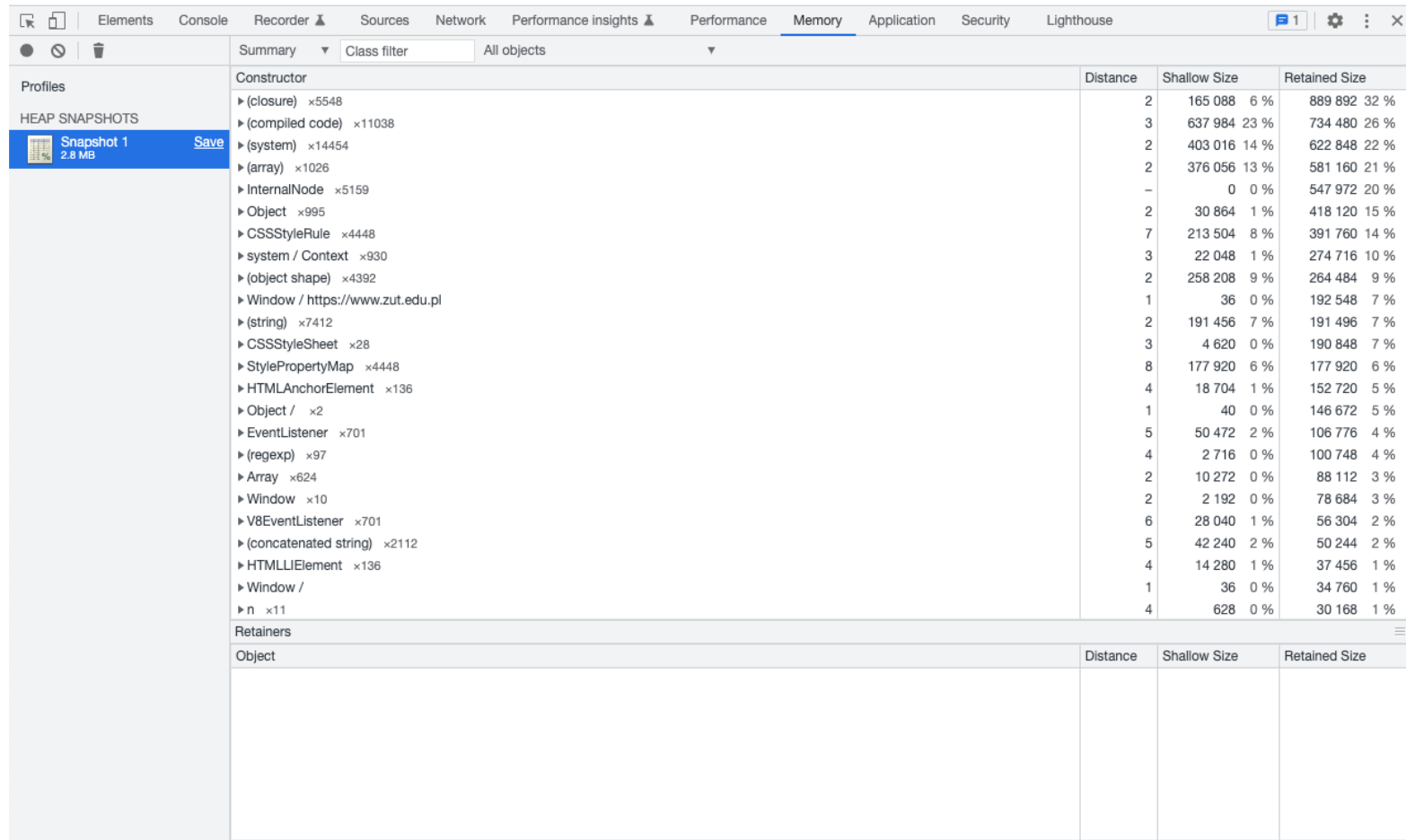
Performance – Speed of loading



Performance – Speed of rendering



Performance – Memory usage

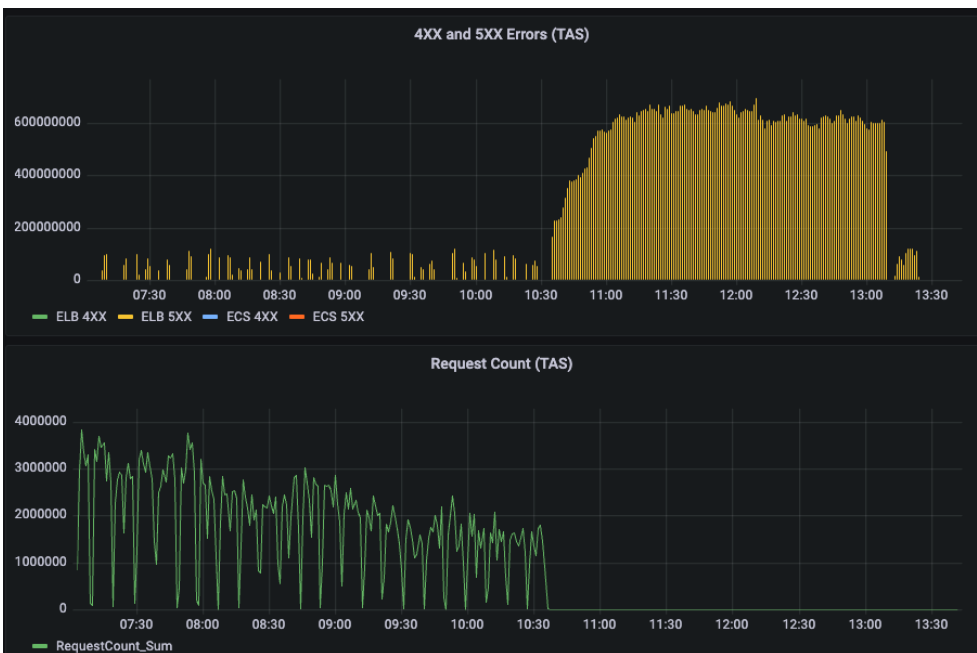


The screenshot shows the Chrome DevTools interface with the Memory tab selected. The left sidebar displays 'HEAP SNAPSHOTS' with 'Snapshot 1' (2.8 MB) selected. The main panel shows a list of memory objects under the 'Constructor' column, including '(closure)', '(compiled code)', '(system)', '(array)', 'InternalNode', 'Object', 'CSSStyleRule', 'system / Context', '(object shape)', 'Window / https://www.zut.edu.pl', '(string)', 'CSSStyleSheet', 'StylePropertyMap', 'HTMLAnchorElement', 'Object /', 'EventListener', '(regexp)', 'Array', 'Window', 'V8EventListener', '(concatenated string)', 'HTMLLIElement', 'Window /', and 'n'. The right panel shows a table with columns 'Distance', 'Shallow Size', and 'Retained Size' for each object.

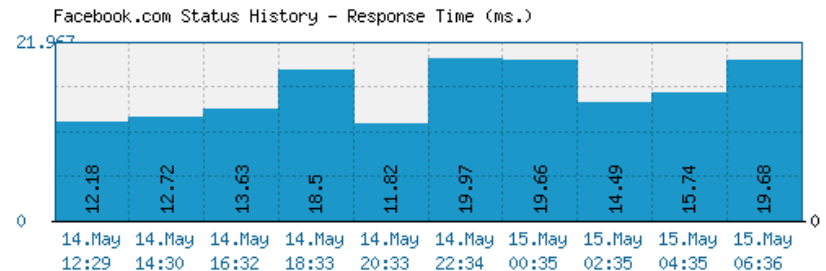
Constructor	Distance	Shallow Size	Retained Size
▶ (closure) x5548	2	165 088 6 %	889 892 32 %
▶ (compiled code) x11038	3	637 984 23 %	734 480 26 %
▶ (system) x14454	2	403 016 14 %	622 848 22 %
▶ (array) x1026	2	376 056 13 %	581 160 21 %
▶ InternalNode x5159	–	0 0 %	547 972 20 %
▶ Object x995	2	30 864 1 %	418 120 15 %
▶ CSSStyleRule x4448	7	213 504 8 %	391 760 14 %
▶ system / Context x930	3	22 048 1 %	274 716 10 %
▶ (object shape) x4392	2	258 208 9 %	264 484 9 %
▶ Window / https://www.zut.edu.pl	1	36 0 %	192 548 7 %
▶ (string) x7412	2	191 456 7 %	191 496 7 %
▶ CSSStyleSheet x28	3	4 620 0 %	190 848 7 %
▶ StylePropertyMap x4448	8	177 920 6 %	177 920 6 %
▶ HTMLAnchorElement x136	4	18 704 1 %	152 720 5 %
▶ Object / x2	1	40 0 %	146 672 5 %
▶ EventListener x701	5	50 472 2 %	106 776 4 %
▶ (regexp) x97	4	2 716 0 %	100 748 4 %
▶ Array x624	2	10 272 0 %	88 112 3 %
▶ Window x10	2	2 192 0 %	78 684 3 %
▶ V8EventListener x701	6	28 040 1 %	56 304 2 %
▶ (concatenated string) x2112	5	42 240 2 %	50 244 2 %
▶ HTMLLIElement x136	4	14 280 1 %	37 456 1 %
▶ Window /	1	36 0 %	34 760 1 %
▶ n x11	4	628 0 %	30 168 1 %

Object	Distance	Shallow Size	Retained Size
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Load testing – stress resilience #1



Facebook Website Status History



The above graph displays service status activity for Facebook.com over the last 10 automatic checks. The blue bar displays the response time, which is better when smaller. If no bar is displayed for a specific time it means that the service was down and the site was offline.

Service Status History

Date	Time	Ping Time	Date	Time	Ping Time
✓ 03.Dec.2024	03:41	13.58 ms.	✓ 03.Dec.2024	05:42	17.16 ms.
✓ 03.Dec.2024	07:43	13.99 ms.	✓ 03.Dec.2024	09:43	18.14 ms.
✓ 03.Dec.2024	11:44	12.41 ms.	✓ 03.Dec.2024	13:45	10.67 ms.
✓ 03.Dec.2024	15:47	14.36 ms.	✓ 03.Dec.2024	17:48	14.78 ms.
✓ 03.Dec.2024	19:48	10.8 ms.	✓ 03.Dec.2024	21:50	15.7 ms.

* Times displayed are PT, Pacific Time (UTC/GMT 0) | Current server time is 23:18

We have tried pinging Facebook website using our server and the website returned the above results. If facebook.com is down for us too there is nothing you can do except waiting. Probably the server is overloaded, down or unreachable because of a network problem, outage or a website maintenance is in progress...

Load testing – stress resilience #2

- **Performance Validation:** Load testing helps determine how a web application performs under normal and peak load conditions. It validates whether the system meets performance requirements and can handle the expected number of concurrent users.
- **Identifying Bottlenecks:** Load testing can reveal bottlenecks in the application, such as slow database queries, inefficient code, or inadequate server resources. Identifying these issues early allows developers to address them before the application goes live.

Load testing – stress resilience #3

- **Ensuring Scalability:** By understanding how the application performs under different loads, organizations can make informed decisions about scaling their infrastructure. This ensures that the system can grow with increasing user demand without compromising performance.
- **Improving User Experience:** A well-performing application leads to a better user experience. Load testing helps ensure that users do not encounter slow response times or crashes during peak usage, which can lead to frustration and abandonment.

Load testing – stress resilience #4

- **Reducing Downtime:** Load testing helps identify potential failure points in the system, allowing teams to implement fixes and optimizations. This proactive approach reduces the likelihood of downtime during high-traffic events, such as product launches or sales.
- **Cost Efficiency:** By identifying performance issues and optimizing resource usage, load testing can help organizations avoid unnecessary costs associated with over-provisioning infrastructure. It ensures that resources are allocated efficiently based on actual usage patterns.

Load testing – stress resilience #4

- **Compliance and SLAs:** Many organizations have service level agreements (SLAs) that specify performance benchmarks. Load testing helps ensure compliance with these agreements by validating that the application meets the required performance standards.
- **Testing Under Realistic Conditions:** Load testing allows teams to simulate real-world user behaviour, including varying load patterns, to understand how the application will perform in different scenarios. This helps in preparing for unexpected spikes in traffic.

Load testing – stress resilience #5

- **Enhancing Security:** Load testing can also serve as a form of stress testing, where the application is pushed beyond its limits to identify vulnerabilities. This helps ensure that the system can withstand malicious attacks or unexpected surges in traffic.
- **Continuous Improvement:** Load testing should be an ongoing part of the development lifecycle. Regular testing allows teams to continuously monitor performance, make improvements, and adapt to changing user needs and technology advancements.

Website improvements #1

Optimize Loading Speed

- Minimize HTTP Requests: Reduce the number of elements on a page (like images, scripts, and stylesheets) to decrease the number of HTTP requests made by the browser.
- Combine Files: Merge multiple CSS and JavaScript files into single files to reduce the number of requests.

Use Content Delivery Networks (CDNs)

- What It Is: CDNs distribute website content across multiple servers located in various geographic locations.
- Benefits: This reduces latency and improves load times for users by serving content from the nearest server.

Image Optimization

- Compression: Use tools to compress images without significant loss of quality. Formats like WebP can provide better compression than traditional formats (JPEG, PNG).
- Responsive Images: Use the srcset attribute to serve different image sizes based on the user's device, ensuring that only appropriately sized images are loaded.

Website improvements #2

Minification and Compression

- Remove unnecessary characters (like whitespace and comments) from CSS, JavaScript, and HTML files to reduce file sizes.
- Enable Gzip compression on the server to compress files before they are sent to the browser, reducing download times.

Leverage Browser Caching

- Set caching rules to store static resources (like images, CSS, and JavaScript) in users' browsers.
- This allows returning visitors to load pages faster, as their browsers can retrieve cached resources instead of downloading them again.

Reduce Server Response Time

- Use efficient server configurations, choose a reliable hosting provider, and consider using technologies like HTTP/2 for faster data transfer.
- Optimize database queries and use caching mechanisms to reduce the time it takes to retrieve data.

Website improvements #3

Implement Lazy Loading

- Load images and other media only when they are about to enter the viewport (i.e., when the user scrolls to them).
- This reduces initial load times and saves bandwidth, especially for pages with many images.

Asynchronous Loading of JavaScript

- Load JavaScript files asynchronously or defer their loading until after the main content has loaded.
- This prevents JavaScript from blocking the rendering of the page, leading to faster perceived load times.

Optimize CSS Delivery

- Inline the CSS required for above-the-fold content directly in the HTML to speed up rendering.
- Load non-essential CSS files after the main content has loaded to improve initial load times.

Website improvements #4

Reduce Redirects

- Minimize the use of redirects, as each redirect creates additional HTTP requests and increases load times.
- Fewer redirects lead to faster page loads and a better user experience.

Monitor Performance

- Utilize performance monitoring tools to analyse website performance and identify areas for improvement.
- Continuously test the website's performance after updates or changes to ensure that optimizations are effective.

Infrastructure improvements

CPU

- Increasing the number of cores or upgrading to a more powerful processor to improve processing power.

RAM

- Adding more memory to allow the server to handle more simultaneous processes and improve performance for memory-intensive applications.

Storage

- Upgrading to larger or faster storage solutions (e.g., SSDs) to improve data access speeds and capacity.

Network Bandwidth

- Increasing the network interface capacity to handle more data transfer and improve connectivity.

Horizontal Scaling

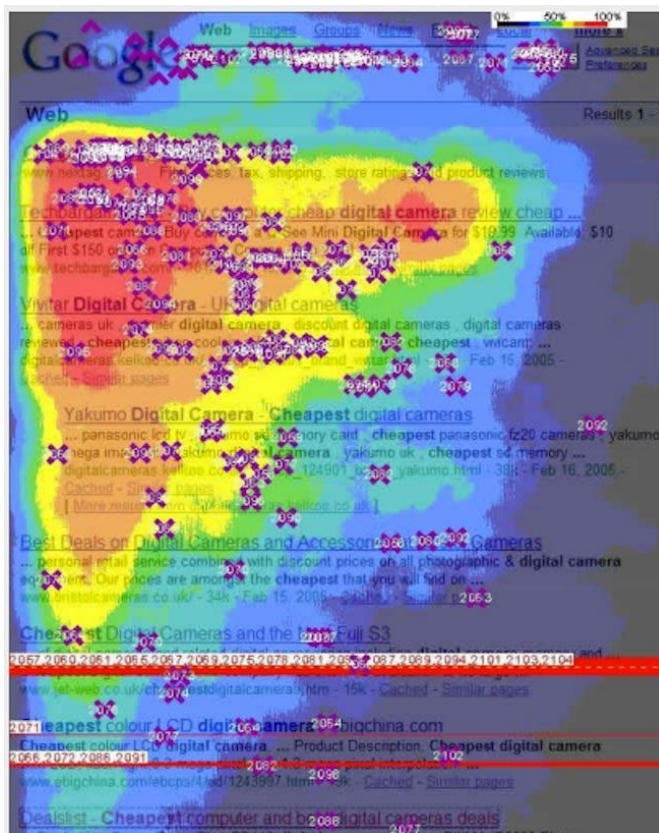
- Horizontal scaling involves adding more servers to distribute the load. This can provide redundancy and improve fault tolerance.

Cloud Solutions

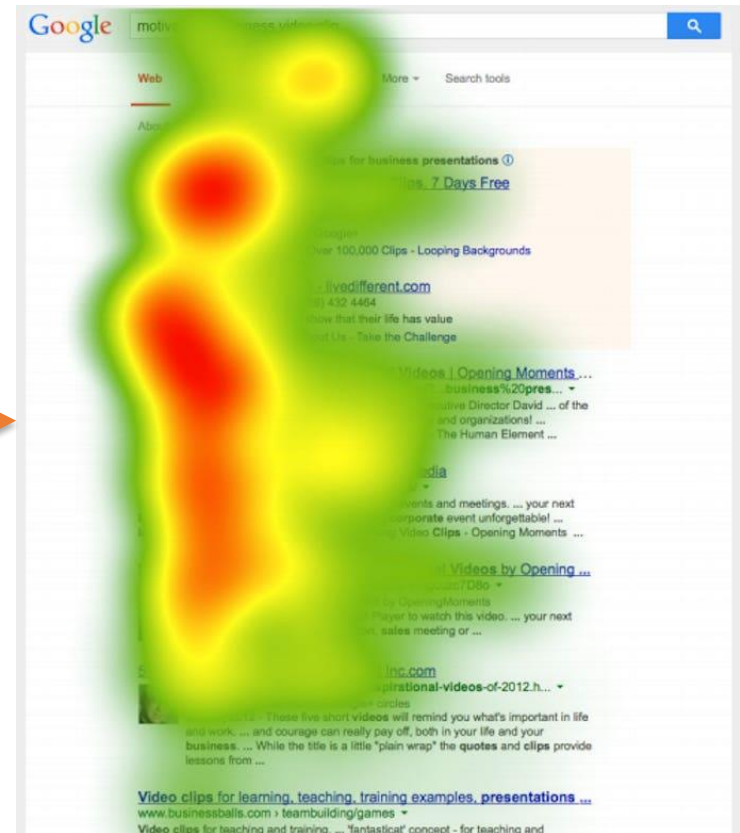
- Utilizing cloud services allows for dynamic scaling, where resources can be adjusted based on demand without the need for physical hardware upgrades.

Observation patterns – Eye tracking

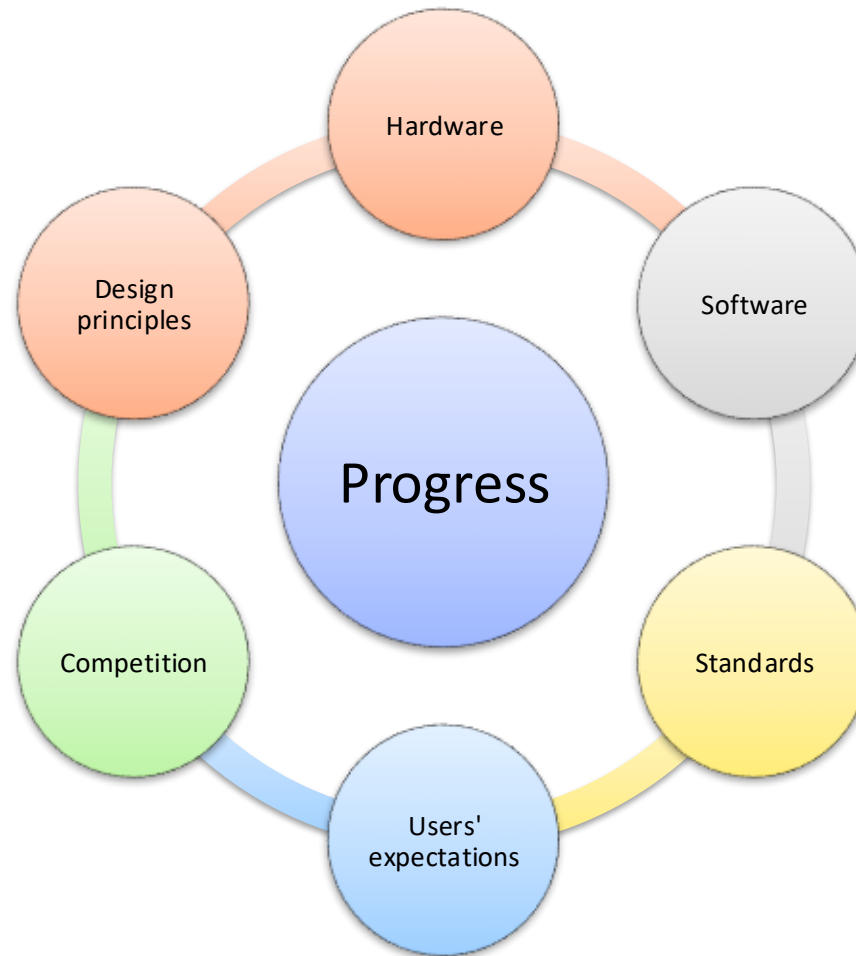
Google – 2005



Google – 2015



Web improvements in time



CSS & HTML Validation

- Validated CSS & HTML means the webpage uses standard styling & code
- **Less problems in the future**
- More coherent rendering across different browsers
- Removal of potential errors, redundant (ineffective) elements
- Potential to automatically process webpages

CSS Validation

Deutsch English Español Français 한국어 Italiano Nederlands 日本語 Polski Português Русский العربية Svenska Български Українська Čeština Romanian Magyar Ελληνικά 繁體中文



The W3C CSS Validation Service

W3C CSS Validator results for <http://www.zut.edu.pl> (CSS level 3 + SVG)

Jump to: [Errors \(5\)](#) [Warnings \(417\)](#) [Validated CSS](#)

W3C CSS Validator results for <http://www.zut.edu.pl> (CSS level 3 + SVG)

Sorry! We found the following errors (5)

URI : <https://www.zut.edu.pl/fileadmin/templates/v6/css/2018-03-20/zut.css?1538639309>

2	Parse Error: only screen and (-webkit-min-device-pixel-ratio:2), only screen and (max-width:1200px) and only screen and (min-resolution:1200px), only screen and (max-width:1200px) and only screen and (min-resolution:192dpi) (.zut-logo-main-wrapper .symbol-pl{background-image:url(../assets/logo-assets/godlo@2x.png);background-size:32px 38px});
2	Parse Error: only screen and (-webkit-min-device-pixel-ratio:2), only screen and (max-width:1200px) and only screen and (min-resolution:1200px), only screen and (max-width:1200px) and only screen and (min-resolution:192dpi) (.zut-logo-main .symbol-pl{background-image:url(../assets/logo-assets/godlo@2x.png);background-size:32px 38px});
2	Parse Error: only screen and (-webkit-min-device-pixel-ratio:2), only screen and (max-width:767px) and only screen and (min-resolution:1200px), only screen and (max-width:767px) and only screen and (min-resolution:192dpi) (.zut-logo-mobile .symbol-pl{background-image:url(../assets/logo-assets/godlo@2x.png);background-size:32px 38px});
2	Parse Error: only screen and (-webkit-min-device-pixel-ratio:2), only screen and (min-width:480px) and (max-width:767px) and only screen and (min-resolution:1200px), only screen and (min-width:480px) and (max-width:767px) and only screen and (min-resolution:192dpi) (.zut-logo-mobile .symbol-pl{background-image:url(../assets/logo-assets/godlo@2x.png);background-size:32px 38px});
64	Value Error : background-image Missing a semicolon before the property name 'width'.

[↑ TOP](#)



Interested in understanding what new technologies are coming out of W3C? Follow [@w3cdevs on Twitter](#) to keep track of what the future looks like!

[Donate](#) and help us build better tools for a better web.

Warnings (417)

URI : <https://www.zut.edu.pl/fileadmin/templates/v6/assets/fancybox/jquery.fancybox.css?1479906092>

32	--webkit-border-radius is a vendor extension
33	--moz-border-radius is a vendor extension
42	--webkit-box-shadow is a vendor extension
43	--moz-box-shadow is a vendor extension
56	--webkit-overflow-scrolling is a vendor extension
118	--webkit-tap-highlight-color is a vendor extension
230	--webkit-border-radius is a vendor extension
231	--moz-border-radius is a vendor extension

URI : <https://www.zut.edu.pl/fileadmin/templates/v6/css/2018-03-20/zut.css?1538639309>

HTML validation

Nu Html Checker

This tool is an ongoing experiment in better HTML checking, and its behavior remains subject to change

Showing results for <https://www.zut.edu.pl/EN/university.html>

Checker Input

Show ☐ source ☐ outline ☐ image report

Check by

<https://www.zut.edu.pl/EN/university.html>

Use the Message Filtering button below to hide/show particular messages, and to see total counts of errors and warnings.

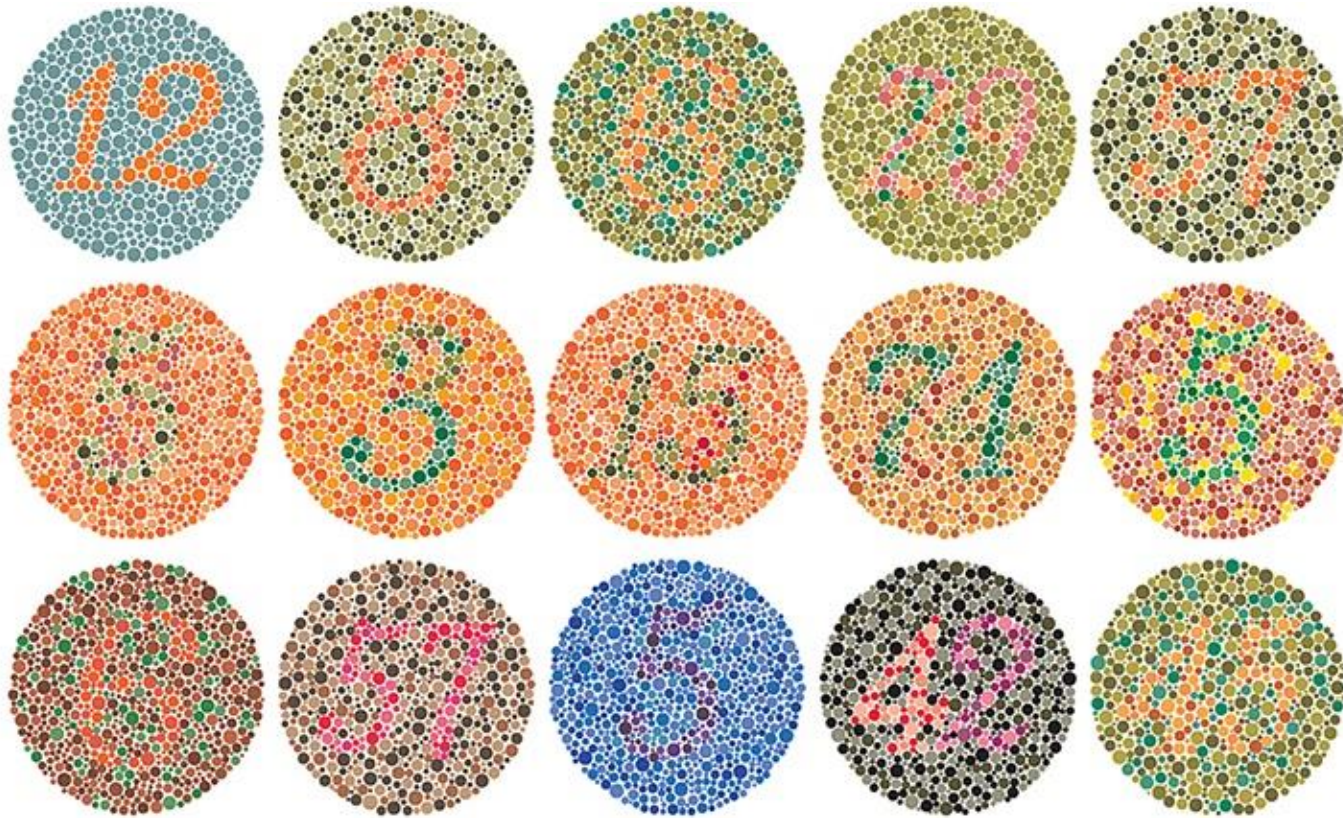
1. **Warning** The `type` attribute is unnecessary for JavaScript resources.
From line 32, column 1; to line 32, column 108
`"all"><script src="/fileadmin/templates/v6/js/base/vendor/jquery-1.10.1.min.js?1379555245" type="text/javascript"></scri`
2. **Warning** The `type` attribute is unnecessary for JavaScript resources.
From line 33, column 1; to line 33, column 105
`</script><script src="/fileadmin/templates/v6/js/base/plugins/jquery.cookie.js?1444996862" type="text/javascript"></scri`
3. **Warning** The `type` attribute is unnecessary for JavaScript resources.
From line 34, column 1; to line 34, column 105
`</script><script src="/fileadmin/templates/v6/js/base/plugins/bootstrap.min.js?1448361424" type="text/javascript"></scri`
4. **Warning** The `type` attribute is unnecessary for JavaScript resources.
From line 35, column 1; to line 35, column 112
`</script><script src="/fileadmin/templates/v6/js/base/plugins/bootstrap.file-input.js?1399220502" type="text/javascript"></scri`
5. **Warning** The `type` attribute is unnecessary for JavaScript resources.
From line 36, column 1; to line 36, column 92
`</script><script src="/fileadmin/templates/v6/js/base/zut-base.js?1630912925" type="text/javascript"></scri`
6. **Warning** The `type` attribute is unnecessary for JavaScript resources.
From line 37, column 1; to line 37, column 106
`</script><script src="/fileadmin/templates/v6/js/scrollmagic/ScrollMagic.min.js?1454827870" type="text/javascript"></scri`
7. **Warning** The `type` attribute is unnecessary for JavaScript resources.
From line 38, column 1; to line 38, column 109

EVALUATION – SPECIAL NEEDS

Colours



Colours – perception



Info about colour blindness

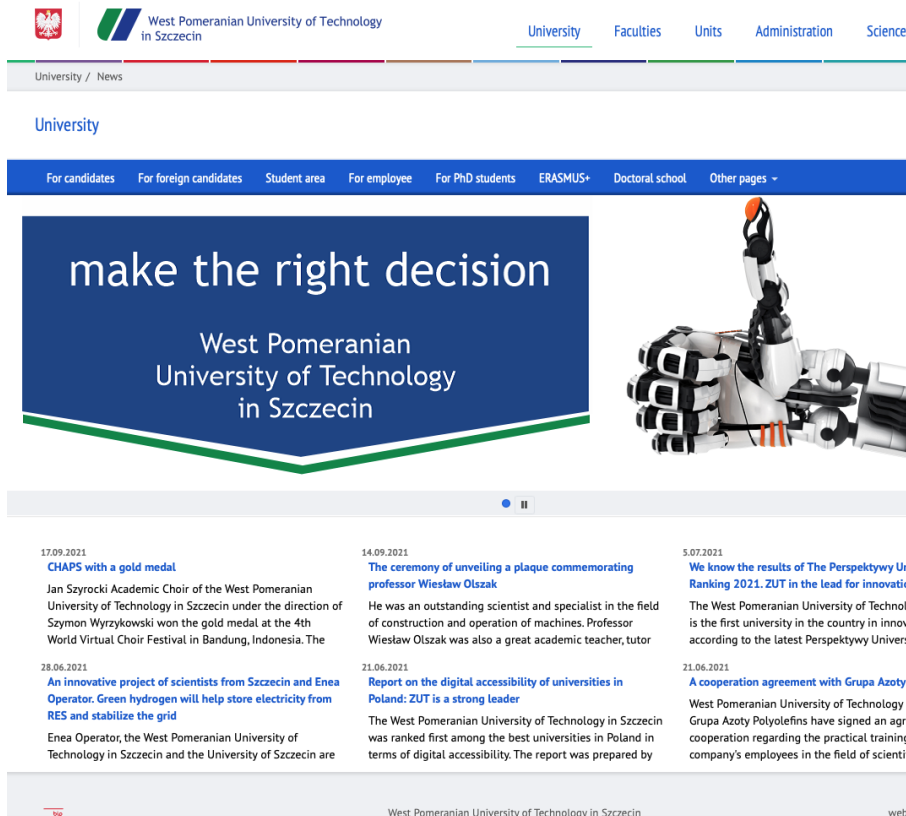
The most common types of colour blindness (or colour vision deficiency i.e. CVD) include:

- Red-green colour blindness: Difficulty distinguishing between reds, greens, and browns.
- Blue-yellow colour blindness: Difficulty distinguishing between blues and yellows.
- Total colour blindness: Rare condition where individuals see no colour at all.

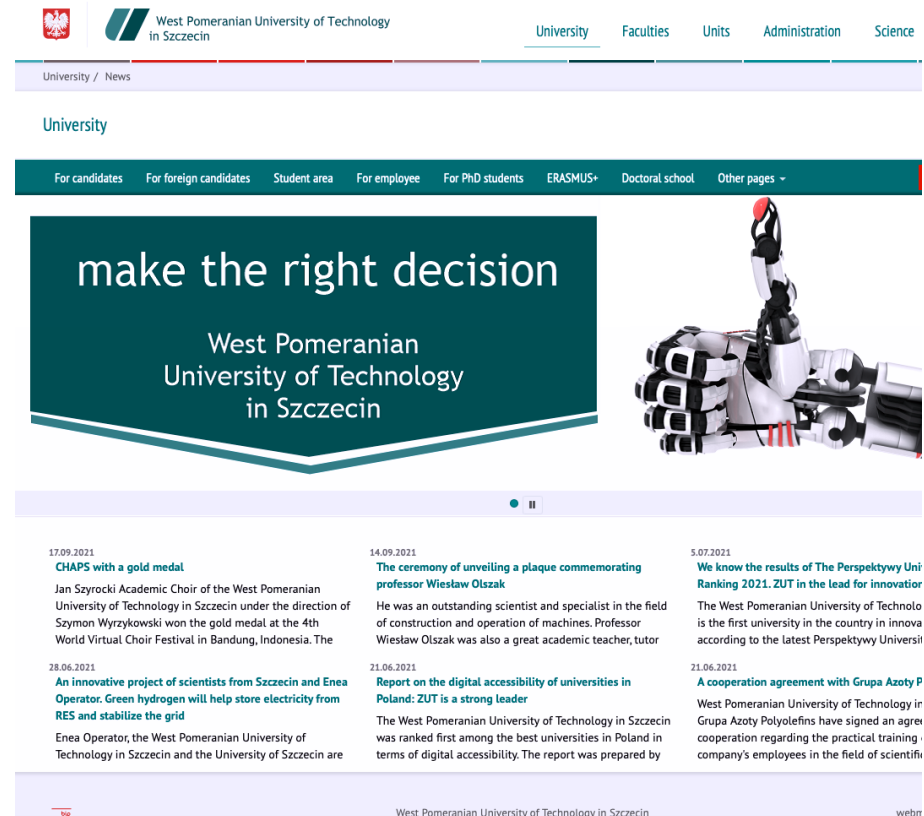
Approximately 8% of men and 0.5% of women of Northern European descent are affected.

Colour Blindness - Example

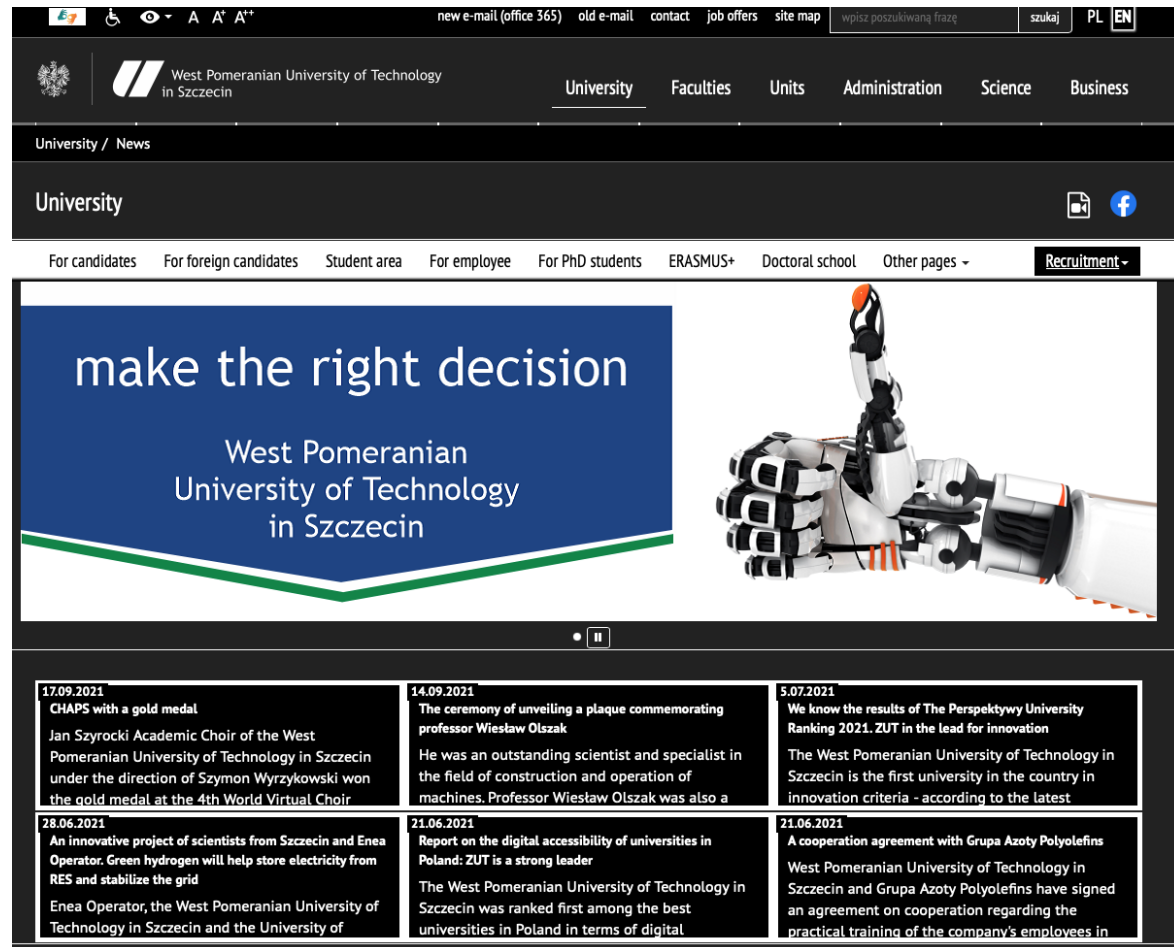
Original



Simulated Blue-Blind/Tritanopia



High Contrast - Example



Needs of (almost) blind users #1

Semantic HTML

- Utilize semantic HTML elements (e.g., <header>, <nav>, <main>, <footer>, <article>, <section>) to provide structure and meaning to the content. This helps screen readers interpret the layout and hierarchy of the page.
- Use heading tags (<h1>, <h2>, etc.) appropriately to create a clear content hierarchy. This allows users to navigate sections easily.

Alternative Text for Images

- Provide descriptive alternative text (alt text) for all images. This text should convey the purpose or content of the image, allowing screen reader users to understand what the image represents.
- For purely decorative images, use empty alt attributes (alt="") to indicate that they do not convey meaningful content.

Keyboard Navigation

- Design the website so that all interactive elements (links, buttons, forms) can be accessed and operated using a keyboard alone. This is crucial for users who cannot use a mouse.
- Provide clear visual focus indicators (such as outlines or highlights) for interactive elements when they are focused using the keyboard.

Descriptive Links

- Use descriptive and meaningful text for links that clearly indicates the destination or action (e.g., "Read our accessibility policy" instead of "Click here"). This helps users understand where the link will take them.

Needs of (almost) blind users #2

Form Accessibility

- Ensure that all form fields have associated labels using the <label> element. This allows screen readers to announce the purpose of each field.
- Provide clear error messages and instructions for correcting mistakes in forms. Use ARIA (Accessible Rich Internet Applications) attributes to enhance error handling.

Use ARIA Landmarks

- Implement ARIA roles and properties to enhance accessibility. For example, use role="navigation" for navigation menus and role="main" for the main content area. This helps screen readers identify different sections of the page.

Readable and Clear Content

- Use clear and concise language to make content easily understandable. Avoid jargon and complex sentences.
- Maintain a consistent layout and navigation structure throughout the site to help users predict where to find information.

Testing with Screen Readers

- Conduct usability testing with blind users who use screen readers to gather feedback on the accessibility of the website. This can provide valuable insights into real-world usage and areas for improvement.
- Test the website with popular screen readers to ensure that all content is accessible, and functions as intended.

Summary

- Proper web design is essential to deliver quality of websites/apps
- User experience influences overall perception of interaction with web solutions
- Do not sacrifice one to gain the other, try to maintain balanced approach
- Evaluate the web systems – take into account its performance and standardisation

TILL NEXT LECTURE